

PICTOS Tutorial

Version 2.0 – 27 October 2026

A step-by-step guide for professionals who support autistic people and others with complex communication needs.

Before you start

PICTOS.net is a tool for making tailor-made pictograms for specific communication contexts, whether for independent living, education or clinical work, without installing anything.

A **pictogram** is a simple drawing that stands for an idea, an action or an object, and helps someone communicate when they do not use (or do not always use) spoken or written words.

PICTOS.net works in your **web browser**, the program you use to go on the internet (Chrome, Safari or Firefox). Everything you do is saved on the same computer you work on. **PICTOS.net** always leaves you in control and respects your privacy.

To begin, open your browser and type **pictos.net** into the address bar at the top.

There is nothing to download or install. To create new pictograms, you will be asked to sign in with an email address or using the 'Continue with Google' button (recommended). This is only a quick check to ensure the artificial intelligence service that draws the pictograms cannot be used by automated programs. You do not need to sign in to view the example libraries or to build sequences and boards from existing pictograms.[1]

If this is your first time, it helps to read this tutorial with **PICTOS.net** open alongside, so you can try each step as you go.

[1]Each person can create up to 50 new images a day.

The analysis stages (Understand and Compose) do not count towards that limit.

What **PICTOS.net** does, in one sentence

PICTOS.net turns a written phrase into a pictogram.

The phrase can be a single word (‘**water**’), a short sentence (‘**I need the toilet**’) or a longer instruction, like this one, which we will use as our example:

> **Look both ways before you cross the road, to make sure no cars are coming**

The tool reads the phrase, decides what should appear in the drawing and proposes a pictogram. That pictogram is saved in your **library**, which is your own collection of pictograms.

You decide whether to accept the pictogram, correct it or throw it away. The artificial intelligence is there to serve your professional judgement, not the other way round.

What you can do with **PICTOS.net**

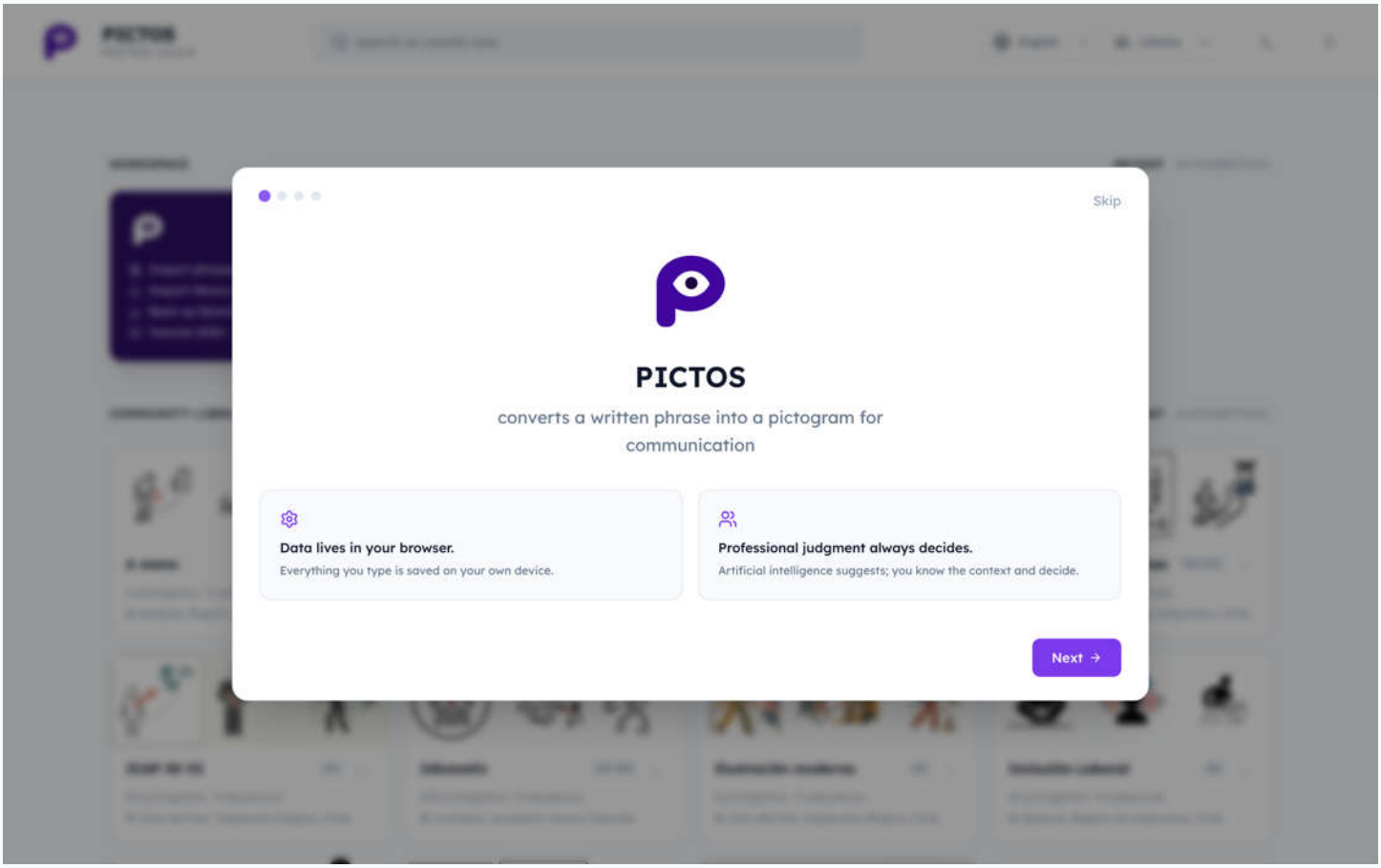
These are the four main things the tool lets you do. Each has its own section in this tutorial.

- **Create pictograms from phrases.** You type what you want to say and **PICTOS.net** draws it.
- **Keep your pictograms in libraries.** You can have several libraries, for example one for each person you support or one per topic, each with its own drawing style.
- **Build sequences.** You put pictograms in order, step by step, to help someone know what is coming: a visit to the GP, the morning routine, an errand. They print with each step numbered.
- **Build communication boards.** A grid of pictograms, coloured by type of word, that the person can point to or tap. Each square can play the recorded voice of someone familiar.

You can also print everything on sheets ready to cut out and laminate.

The home screen

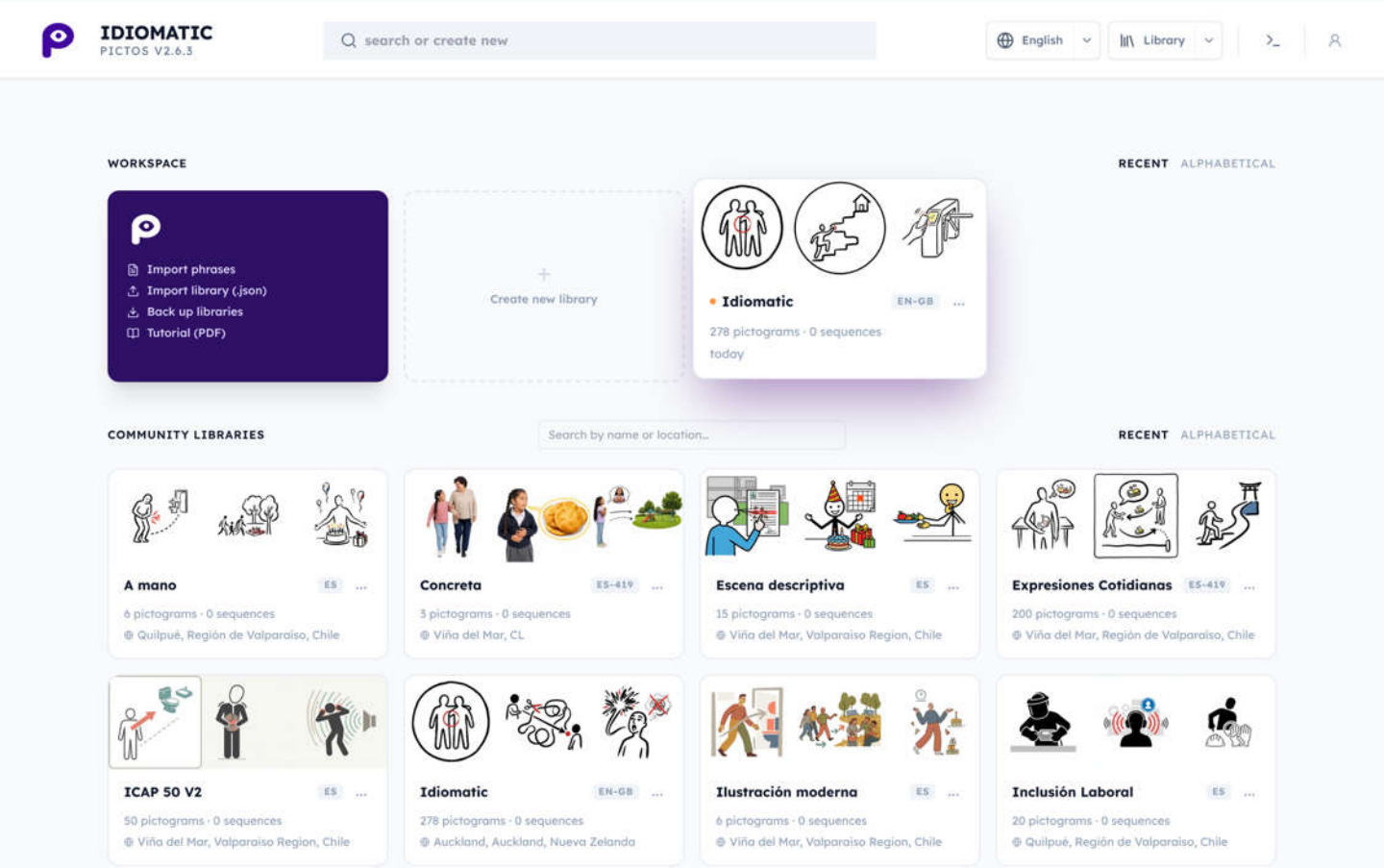
The first time you open **PICTOS.net**, a short four-screen welcome appears. Move on with the ‘Next’ button or close it with ‘Skip’.



Welcome: your data lives in your browser, and professional judgement always decides.

After that you will see the home screen, which has two areas. At the top is your **Workspace**, where your own libraries appear, next to a purple panel of shortcuts. Below

it are the **Community libraries**: example collections made by other people in different drawing styles. You can search them by name or by place.



The Workspace with one library of your own and, below, the community libraries.

To start with an example, click the three dots '…' on a community library and choose **Open**. **PICTOS.net** makes a copy in your Workspace; anything you change there leaves the original untouched. In this tutorial we use the Idiomatic library, which has 278 everyday English expressions, from 'Make a cuppa' to 'Over the moon'.[2]

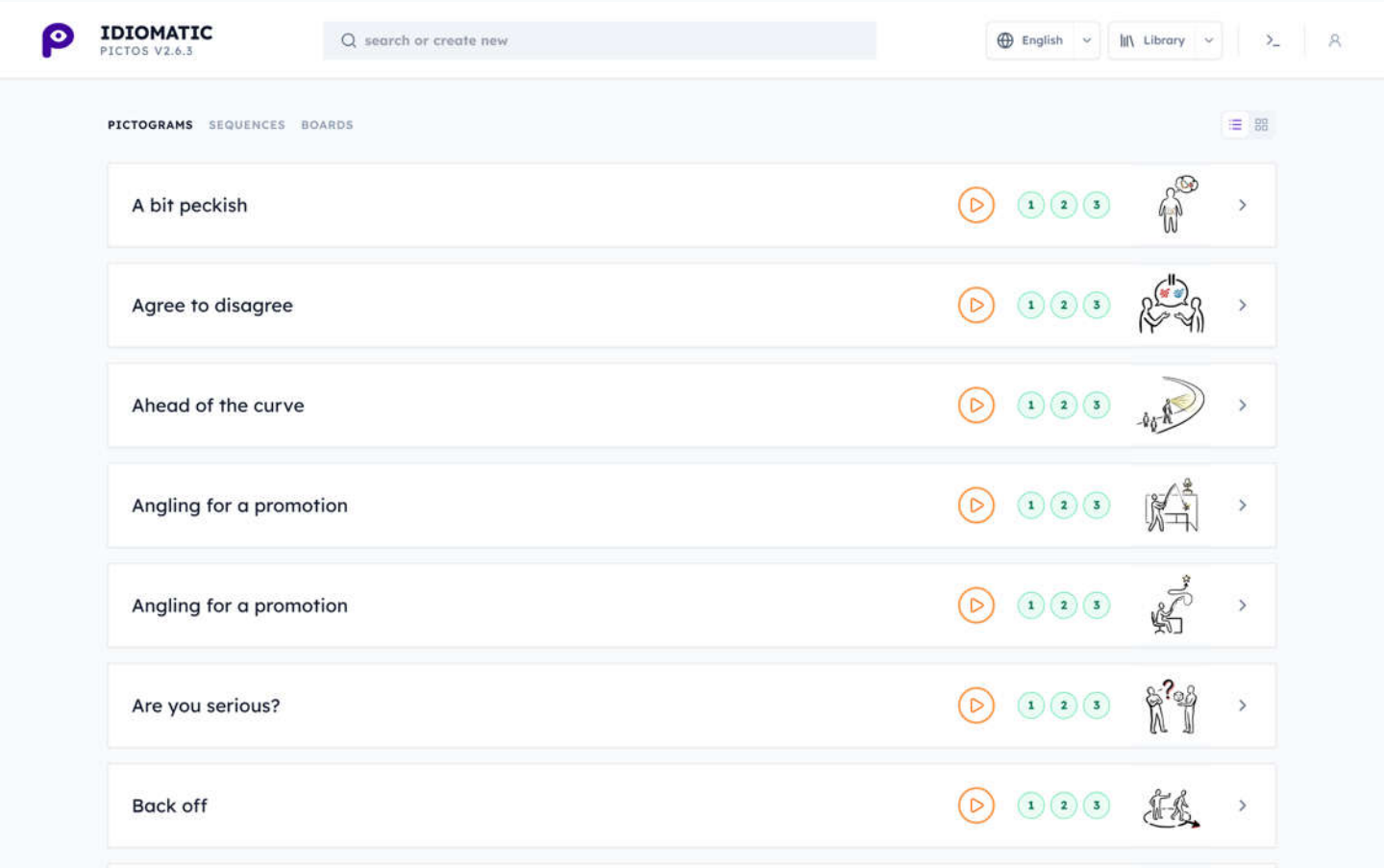
If you would rather start from scratch, click **Create new library** and give it a name, for example Aroha's pictograms or Te Awa School.

[2]Idiomatic is the largest English library available, which is why we use it here. Bear in mind that idioms can be hard for people who read language literally, as many autistic people do: on a real board, plain and literal phrases are usually the better choice.

Inside a library

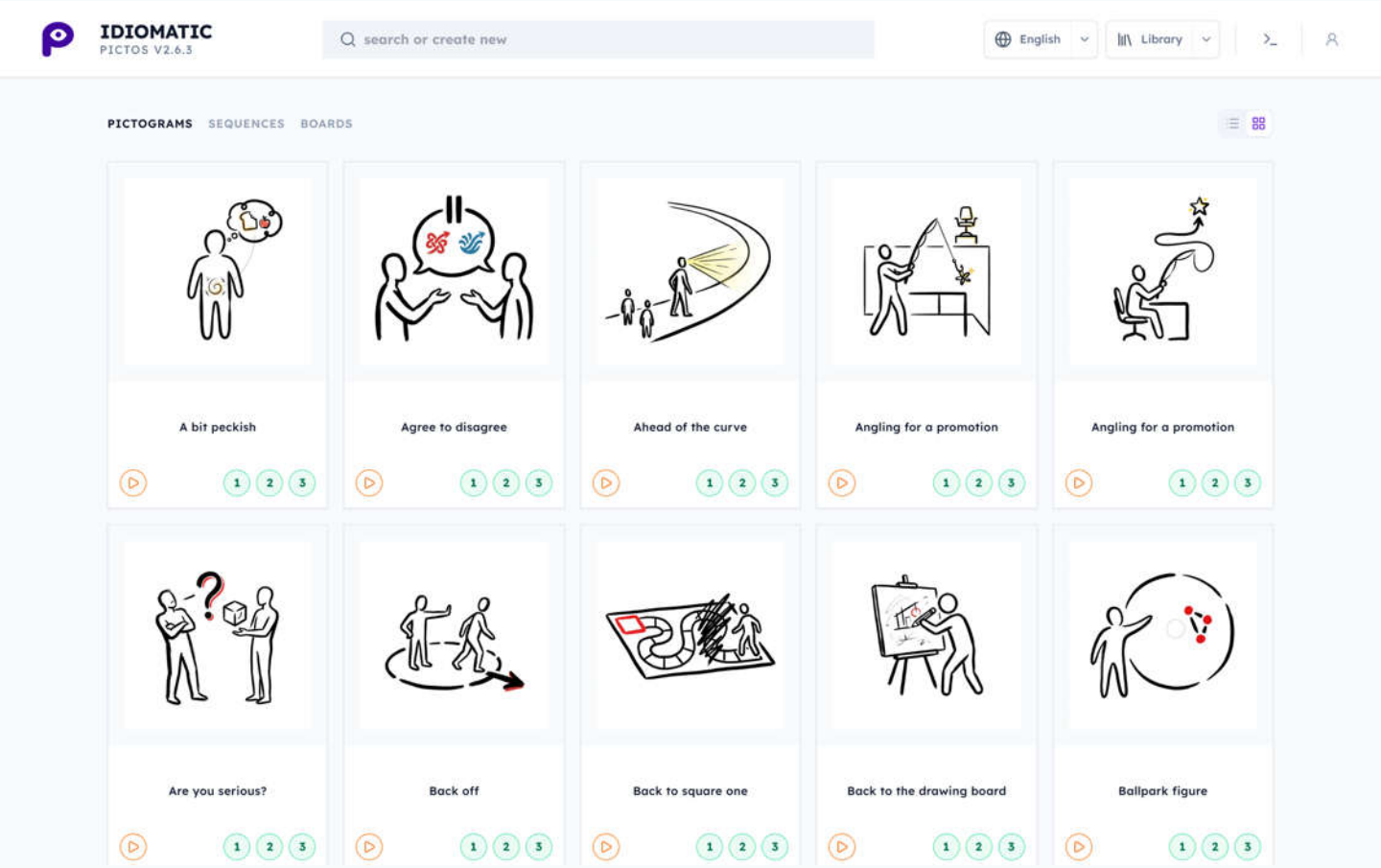
When a library is open, its name appears at the top left. Underneath are three **tabs**, like the tabs on a folder: **Pictograms**, **Sequences** and **Boards**. Click one to go to that part. You always land on Pictograms.

Pictograms can be shown in two ways. To switch, use the two small icons on the right. The **list** shows one row per phrase, with its three stages and the finished drawing. This is the view for working: correcting, redrawing, checking.



List view: each row shows the phrase, its three stages and the final image.

The **grid** shows the drawings large, with the phrase underneath. This is the view for looking at the whole set: whether the style is consistent, whether a drawing looks out of place, whether something is missing.



Grid view: useful for reviewing the whole set at a glance.

Creating your first pictogram

Everything starts in the search box, at the top centre of the screen.



1. Click the box that says ‘search or create new’.
2. Type the example phrase:

   Look both ways before you cross the road, to make sure no cars are coming

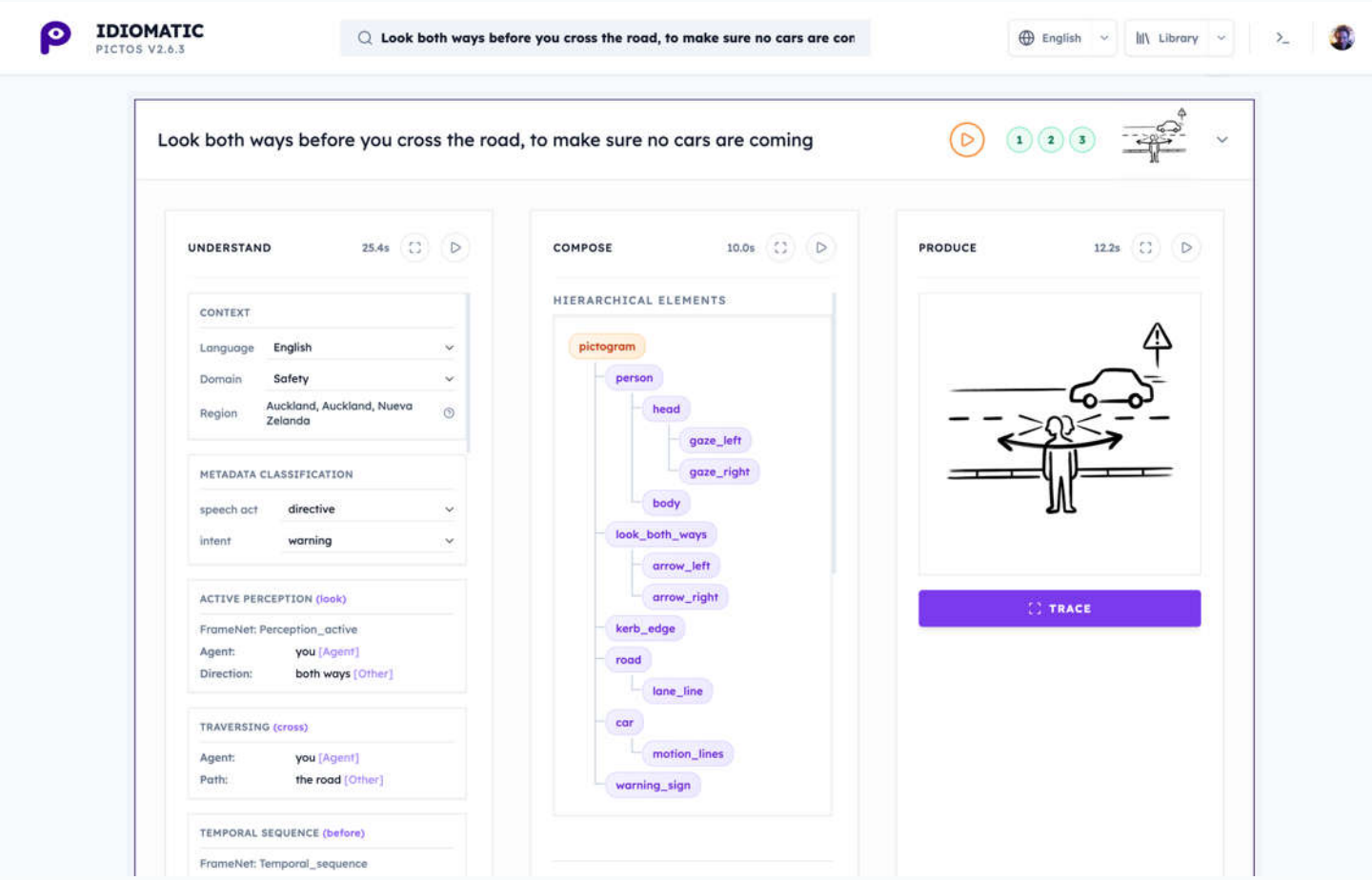
3. Press the **Enter** key (the large key on the right of the keyboard, the one you use to start a new line).

A new row appears with three numbered circles: these are the three stages **PICTOS.net** goes through to make the drawing. When all three turn green, the pictogram is ready. It takes less than a minute.

You can open the row with the arrow on the right to watch it progress. The same box also finds a phrase you already have: if you type it and it appears in the list, no duplicate is made.

The three stages you will see

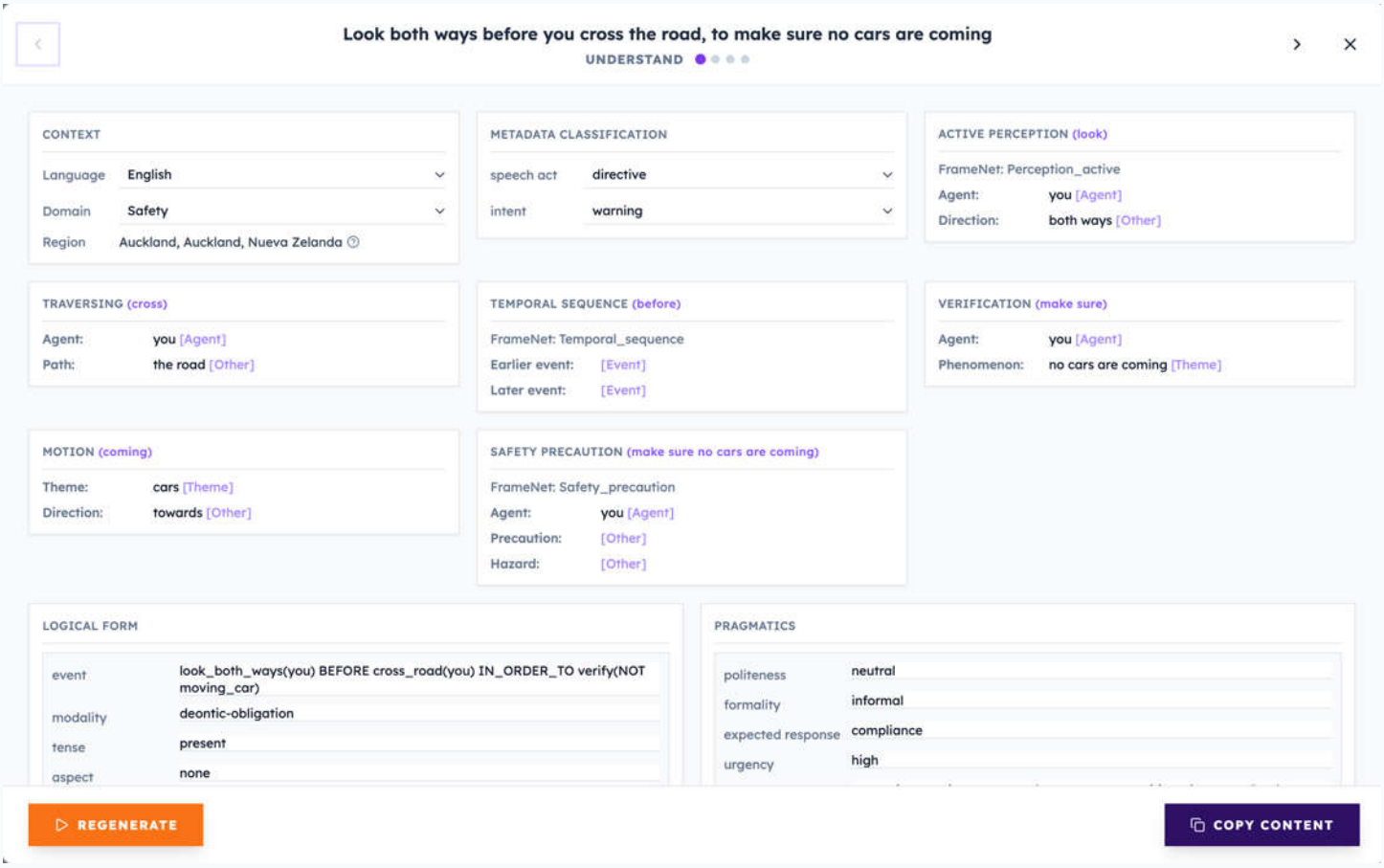
Opening a row shows the three stages side by side. Here is the row for our example phrase. You do not need to understand every detail: it is enough to know what each stage does, and that every one of them can be corrected.



An open row: Understand, Compose and Produce, side by side.

Understand

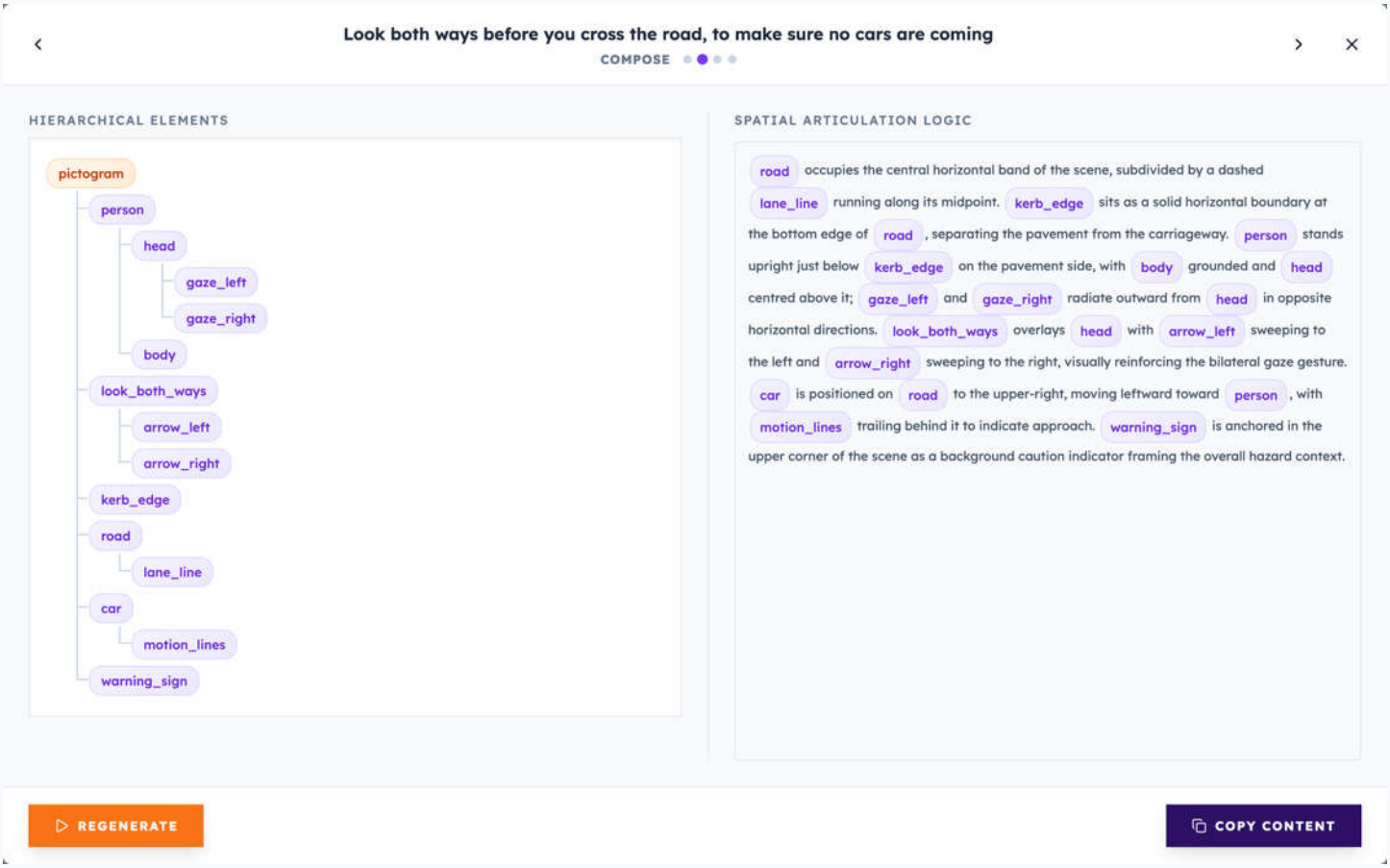
PICTOS.net reads the phrase and works out what it means: who does what, where, and with what intention (asking, refusing, warning, commenting…). It takes a few seconds.



Detail of how the phrase was understood.

Compose

Next, **PICTOS.net** decides which things must appear in the drawing (the person, the road, the car, the look to each side) and where each one goes.



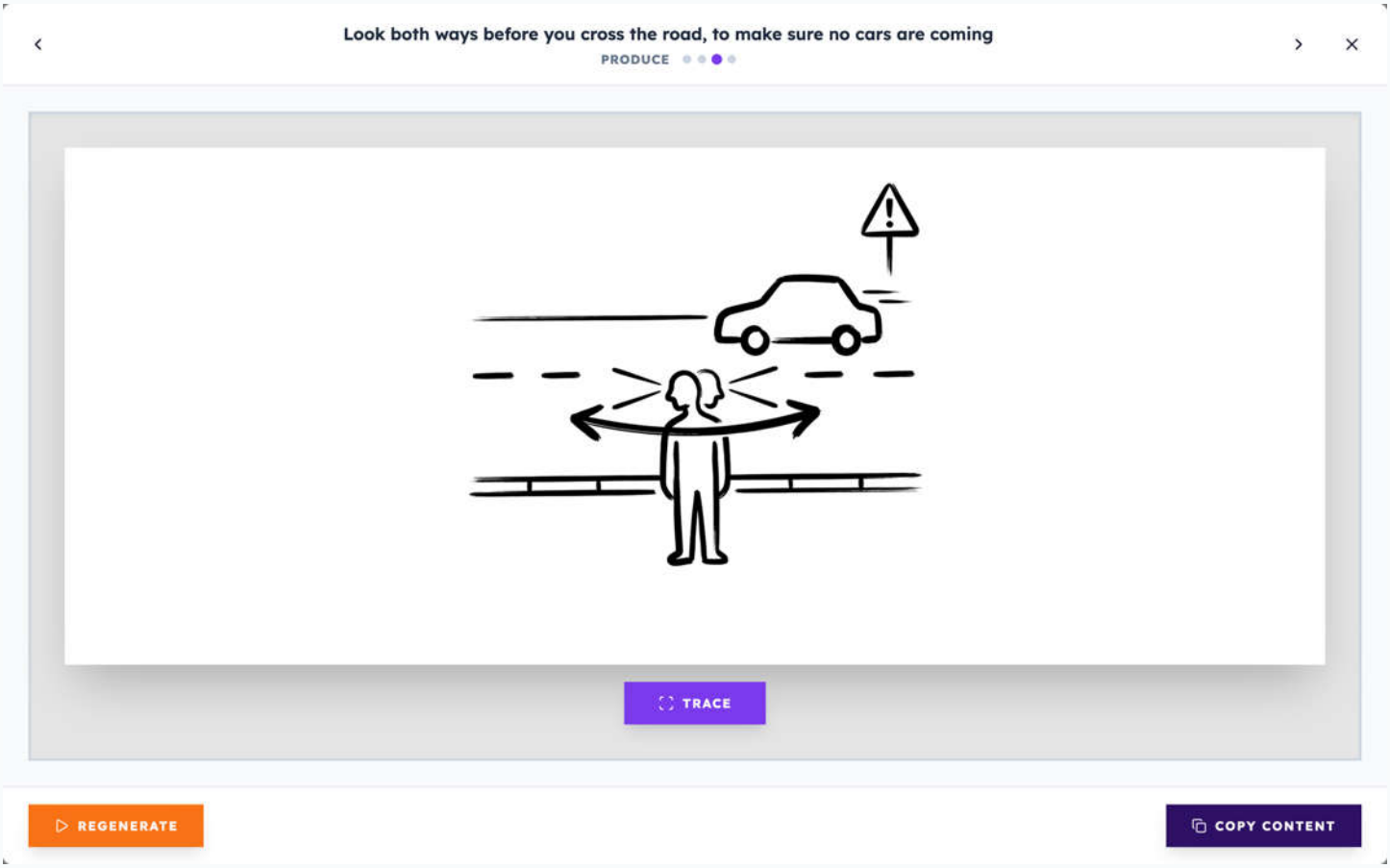
The list of things that make up the drawing.

You can correct that list: (1) **rename** an item, (2) **drag** it with the mouse to another place in the list, or (3) **add** a new one. Afterwards you can ask for the description of how the drawing is laid out to be rewritten automatically.

That description can also be corrected by hand, like any other text. To refer to an item in the list, write it in single quotes ('like this') and **PICTOS.net** will know which one you mean. Then you can redraw the picture in the next stage.

Produce

PICTOS.net draws the pictogram. If you are not happy with it, you can ask for another drawing (this stage only), correct the earlier stages, or delete the row and write the phrase another way.

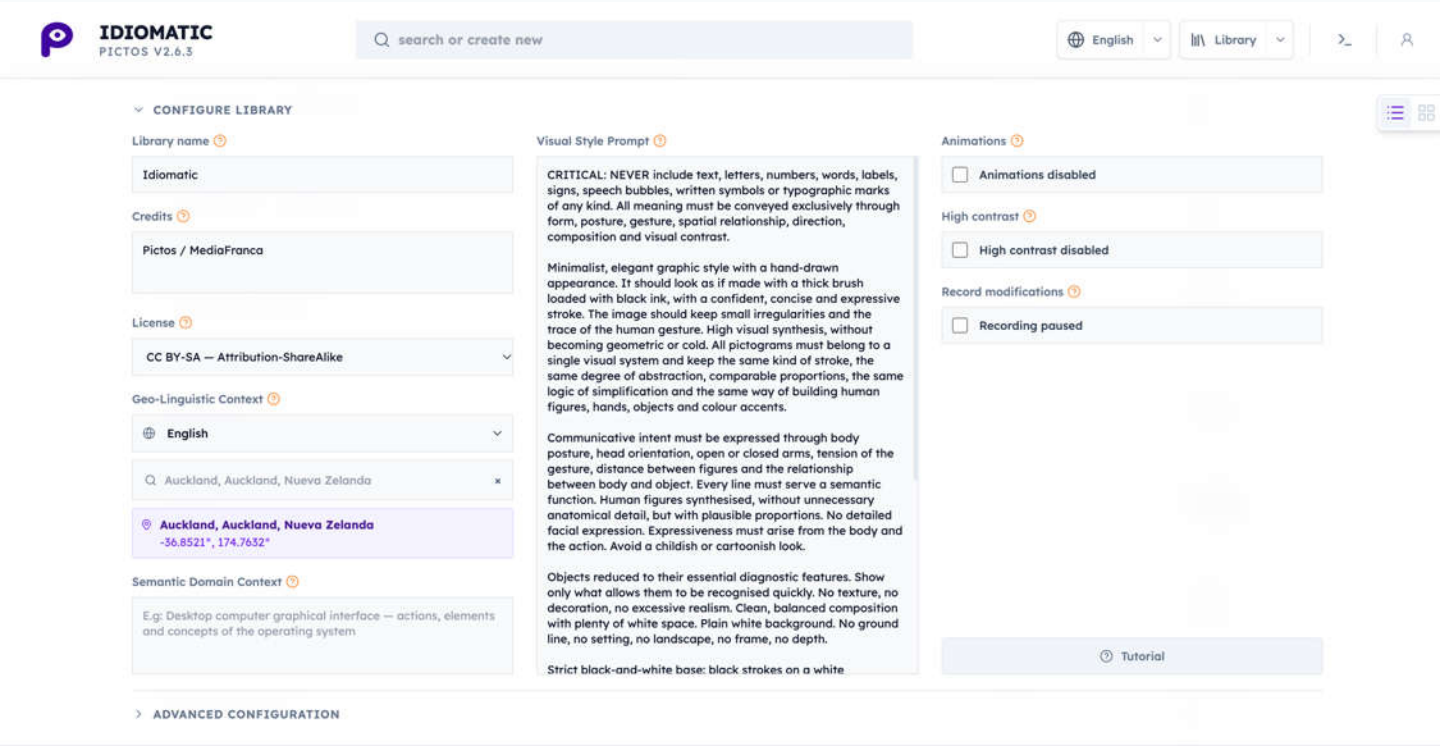


The finished pictogram, ready to download or copy.

Every pictogram is kept in your library, inside your browser.

Configuring the library

Before making lots of pictograms, it pays to set the library up. You only do this once. At the top right is the **Library** button; click the little arrow beside it and choose **'Configure Library'** (the gear icon).



Configure Library: name, context, drawing style and accessibility.

Library name. What the library will be called. For example: Te Awa School or Aroha's pictograms.

Credits. The people or organisation responsible. This goes with the library whenever you share it.

License. Says how other people may use your pictograms[3]. For clinical and educational material, CC BY is usually enough: it lets others use them as long as they credit you. If you prefer to keep all rights, choose copyright.

[3] Only of you explicitly choose to share the library via email with me and ask to publish it as part of the example libraries.

Geo-Linguistic Context. The language and the place where the pictograms will be used. Entering the real town (Dunedin, Rotorua, Whanganui) helps the drawings look like what the person knows: a ‘dairy’ in Aotearoa is not the same as a corner shop in London.

Semantic Domain Context. One line describing the world your phrases belong to. It helps each word be understood in the right sense. For example, if the library is described as Independent living for autistic young people and adults in Aotearoa New Zealand: home, public transport, health and everyday errands, the word ‘take’ is understood as taking the bus or taking your medication, not as grabbing something. In a library about using a computer, ‘window’ and ‘folder’ are understood as things on the screen.

Visual Style Prompt. A written description of how you want every drawing to look. The next section explains how to write one.

Animations and High contrast. Options that make the **PICTOS.net** screen itself easier to use: turning off movement, and making borders and lettering stronger.

Record modifications. When switched on, **PICTOS.net** notes every change you make by hand to a pictogram. Useful if you need to show how you arrived at a version.

Once it is set up, every pictogram you create follows that style and that context. That way the tenth phrase looks just as consistent as the first.

How to write a good visual style prompt

A **prompt** is a written instruction for the artificial intelligence. The style prompt is the most important ‘dial’ in **PICTOS.net**: it decides how every pictogram in a library will look.

Unless told otherwise, the artificial intelligence tends to add details that make a pictogram harder to read: lettering inside the drawing, very expressive faces, shadows, depth, floors, frames and decoration. That is why a good prompt says what you want as well as what you do not.

Six ingredients of a good prompt

1. **A familiar style as a reference.** Describe the kind of drawing you want in everyday words: ‘like airport signs’, ‘flat coloured drawing’, ‘child's pencil drawing’, ‘phone app icon’.
2. **The kind of shapes.** Simple shapes (circles, rectangles, rounded forms), straight lines and gentle curves, or hand-drawn strokes. This decides whether the pictogram feels cooler or warmer.
3. **The colours.** A few colours (three to five), without shading, on a white background. It helps to give each thing its own colour: one for the person, another for the object, another for the action.
4. **The human figure.** This makes the biggest difference to whether the pictogram is understood. Ask for a simple figure, with no detailed face, no fingers and no particular clothing. The drawing has less clutter and does not look like anyone in particular.
5. **The layout.** One idea per pictogram, in the centre, with plenty of white space around it. If the phrase has steps (before, look, make sure), ask for the order to be clear.
6. **What you do not want.** Just as important: no lettering in the drawing, no textures, no shadows, no depth, no floor, no frames, no signature.

A complete example

This prompt brings all six ingredients together and gives pictograms that are easy to read on a communication board. You can copy it straight into the ‘Visual Style Prompt’ box and adjust it as you see the results.

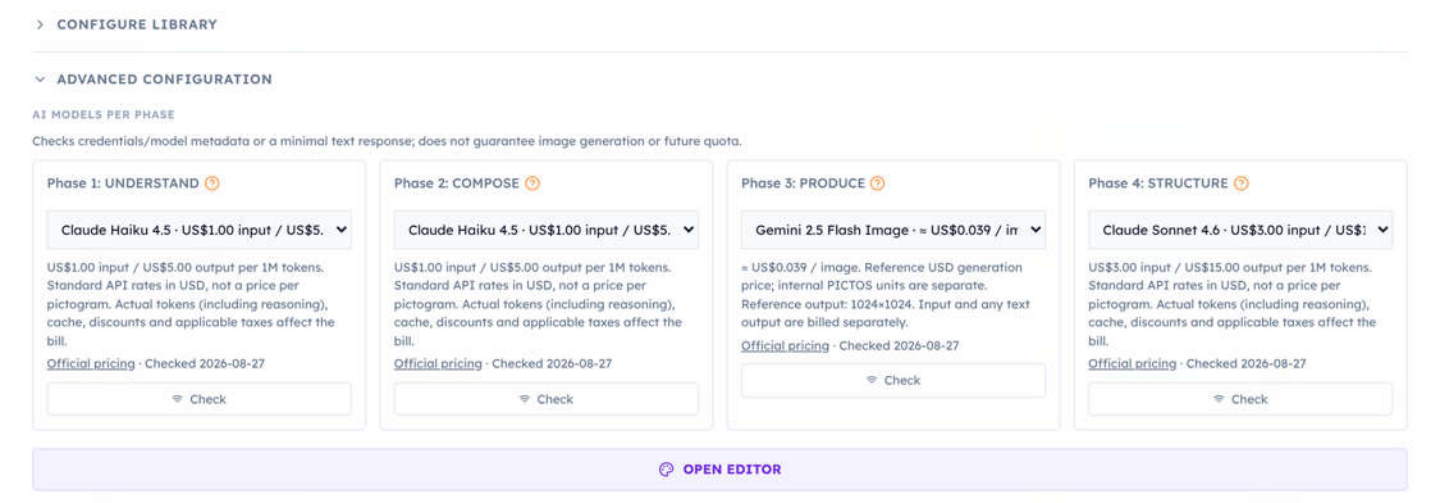
> Public signage style pictogram, with figures and objects built only from geometric primitives (circles, rectangles and rounded shapes), with clean precision and no hand-drawn strokes. Flat, saturated colours, five at most, on a white background, with a different colour for the person, the object and the action. The human figure is an essential synthesis of posture, with no face, no fingers and no specific clothing. Centred composition with generous negative space. Avoid text on the image, avoid textures, avoid shadows, avoid perspective, avoid a floor and avoid decorative elements. A single idea, clear and direct.

To see how much the result depends on the prompt, open two or three community libraries and compare them: similar ideas are drawn in very different ways.

Advanced configuration

You can skip this section. **PICTOS.net** is set up to work well without changing anything here. It is only useful if you want to compare results or if something stops working.

Below ‘Configure Library’ there is a second section, closed at first: **Advanced configuration**. Click its name to open it.



Advanced configuration: which artificial intelligence does each stage.

An artificial intelligence for each stage

Behind **PICTOS.net** there are several artificial intelligence programs, each specialised in one task. This section lets you choose which one does each stage, a bit like choosing which specialist to give each part of a job to:

- **Phase 1, Understand**, and **Phase 2, Compose**: the programs that read the phrase and plan the drawing. They are quick and do not count towards your daily limit.
- **Phase 3, Produce**: the program that draws. It makes the most visible difference to the result. Some give a fixed picture and others a drawing that can be corrected line by line.
- **Phase 4, Structure**: an optional stage that names each part of the drawing (the person, the object, the action). This lets you correct the drawing part by part, and lets screen readers describe it.

Each option shows a **reference price**.[4] The **Check** button tests that the option is working; it does not create any drawing. Options marked ‘experimental’ work, but have not yet been tested enough.

A worked example

Say that in the Idiomatic library you want to see whether another option draws people better. You change the option for **Phase 3: PRODUCE**. Because the library already has 278 pictograms, **PICTOS.net** asks what to do with them:

- **Change only**: the pictograms you already have stay as they are, and only new ones use the other option. This is the cautious way to try it out.
- **Change and regenerate all**: redraws the whole library, and every drawing counts towards your daily limit.

The sensible approach is to choose ‘Change only’, redraw two or three phrases as a test, and compare before deciding.

Style editor

At the end of the section is **Open editor**, where you can change the line thickness, fill colours and outlines of the drawings that can be corrected line by line. It is useful, for example, to switch a whole library to high contrast without redrawing it.

[4]These are the rates charged by the artificial intelligence companies, in US dollars, and are only there to compare options. They are not a charge to you: in PICTOS.net, use is counted in images per day, not in money.

Sequences

A **sequence** is a set of pictograms in order, showing the steps of an activity. It helps someone know what is coming, which many autistic people find reassuring when something is new or out of routine.[5]

Sequences are built from pictograms in the same library. If you delete a sequence, the pictograms stay in your library.

Example: a morning routine

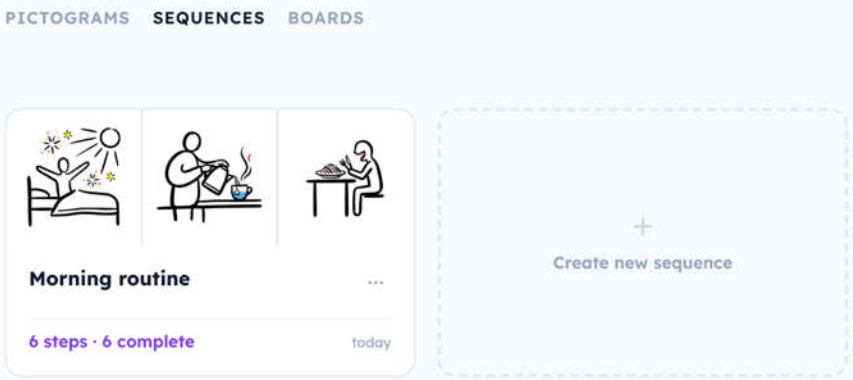
We will build a six-step sequence for someone getting the day started at home:

1. Rise and shine
2. Make a cuppa
3. Tuck in
4. Wash up
5. Tidy up
6. Nip to the shop

[5]Sequences like these are also known as visual schedules or visual scripts.

Create the sequence

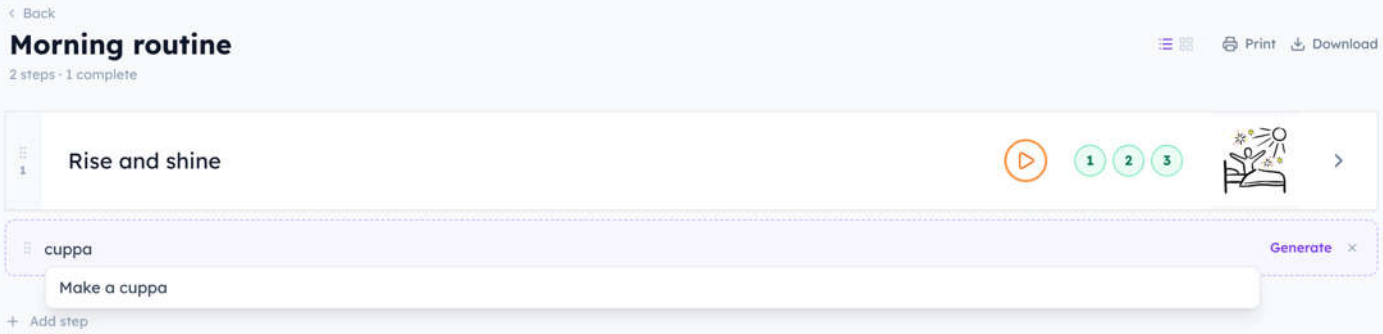
Click the **Sequences** tab and then the **Create new sequence** box. An empty sequence called ‘Untitled sequence’ opens. Click that title, type Morning routine and press Enter.



The library's sequences, with the box for creating a new one.

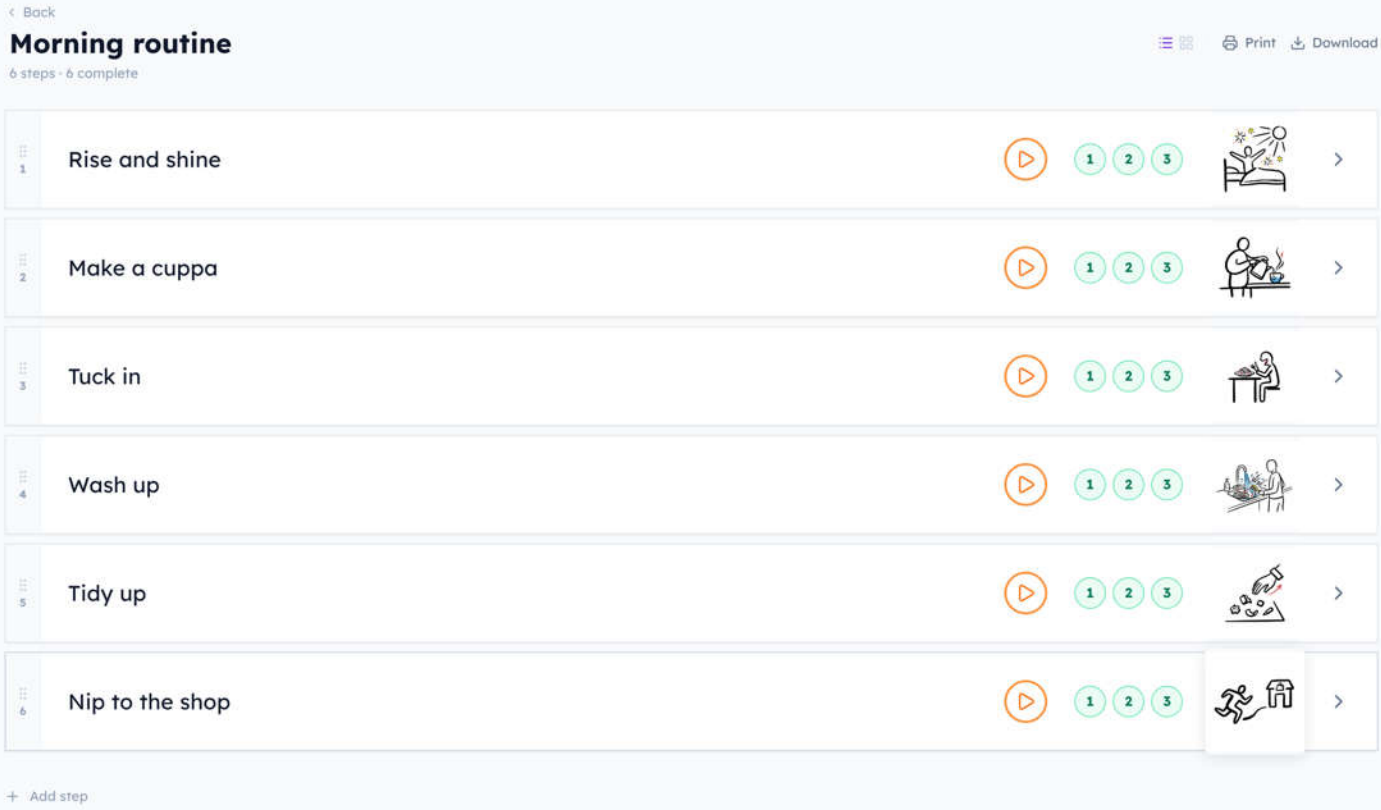
Add steps

Click **Add step** and start typing. You do not need to type the whole phrase: typing ‘cuppa’ brings up Make a cuppa underneath. Click it and the step is done, with its pictogram.



As you type, PICTOS shows matching pictograms from the library.

Repeat for each step. This is the finished sequence:

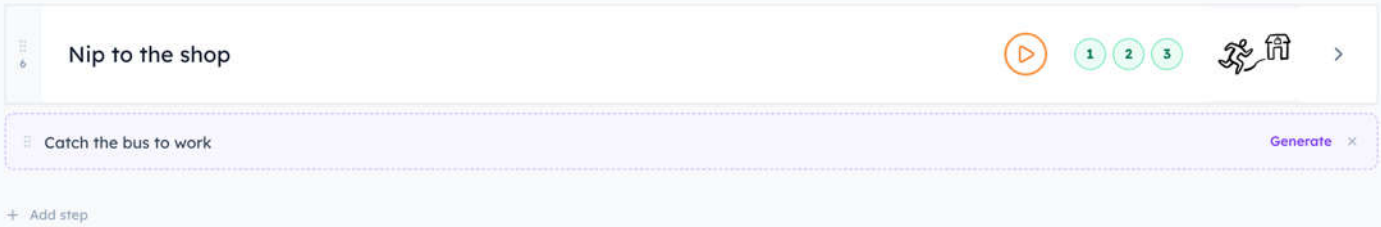


The ‘Morning routine’ sequence with its six steps.

When the pictogram does not exist yet

If you type a step that is not in the library, for example Catch the bus to work, nothing appears underneath and the step gets a dotted border. **Generate** appears on the right:

click it and **PICTOS.net** creates the pictogram, keeps it in the library and puts it in that step.[6]

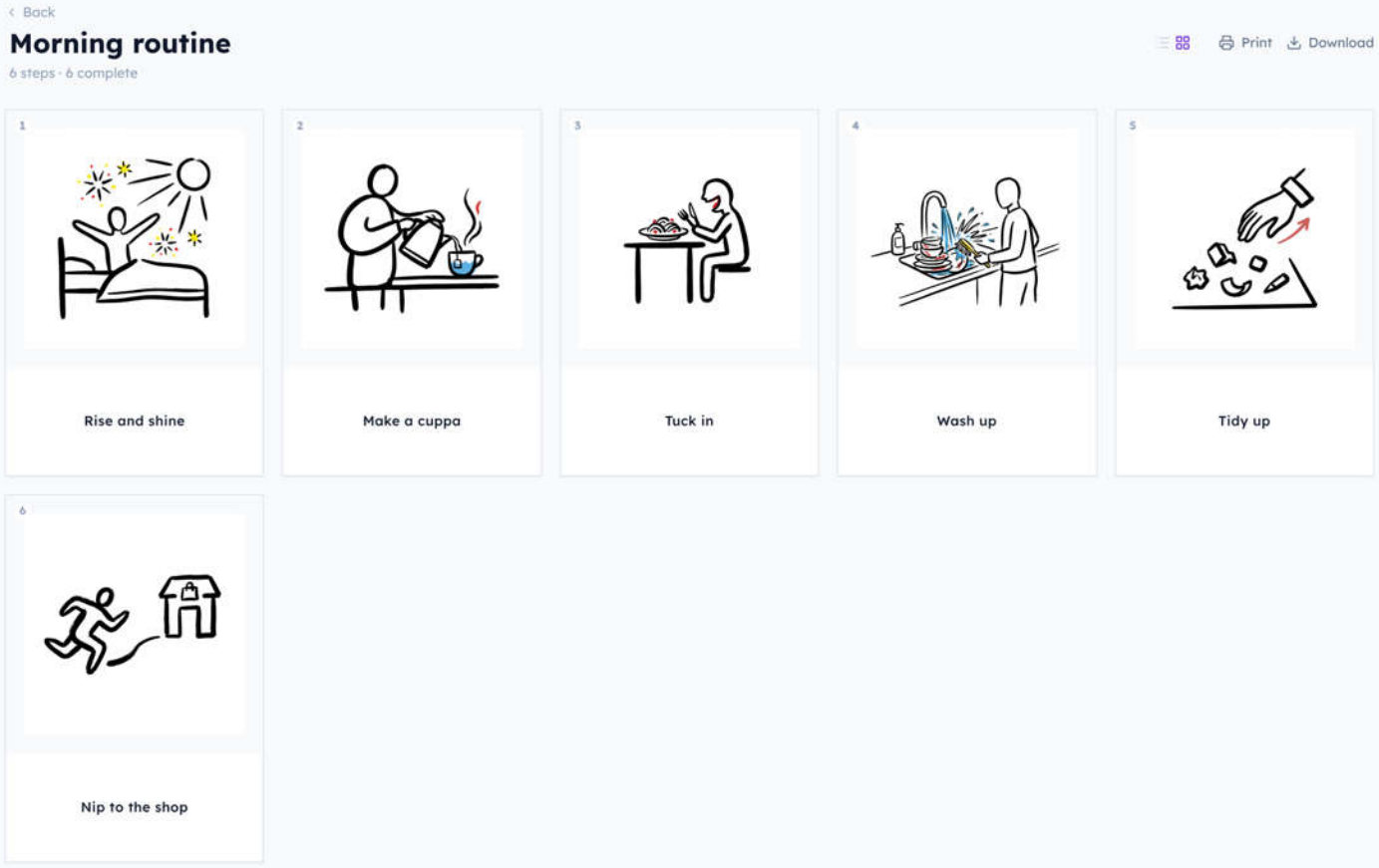


A new step, still without a pictogram, with the Generate button.

[6]Watch out: if you type and press Enter without choosing one of the phrases that appear underneath, a new pictogram is created too. To use one you already have, click the phrase that appears.

Reorder, change and remove steps

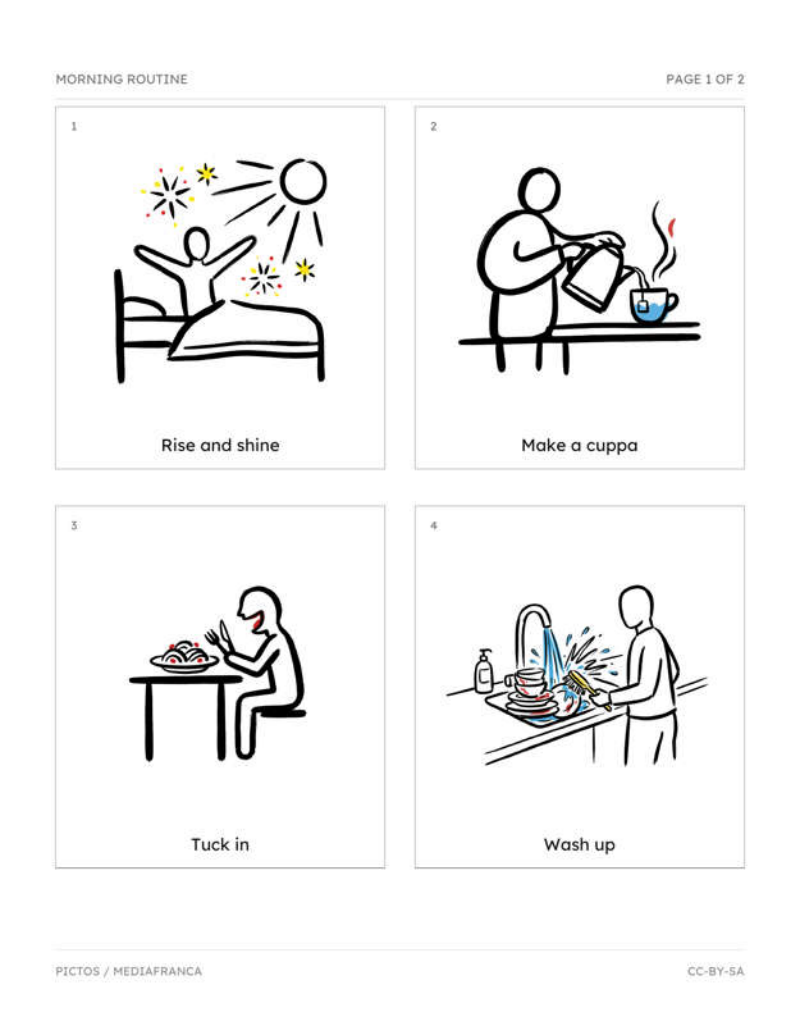
- To **change the order**, hold the mouse button down on the dots to the left of a step and move it to its new place. The numbers update by themselves.
- To **remove** a step, click the cross on the right. The pictogram is not deleted from the library.
- To **see the sequence as cards**, click the squares icon at the top right. Each card shows its step number in the corner.



The sequence shown as cards, with each step's number.

Print and download

Print prepares a document for printing (a PDF) on Letter-size paper, which is very close to A4: the sequence name at the top, four pictograms per sheet, the phrase under each one and its step number. The lines are guides for cutting out and laminating.



The first sheet of the sequence, ready to print.

Download saves the sequence drawings onto your computer, numbered in order (01, 02, 03…), so you can use them in another program or a presentation.

Communication boards

A **communication board** is a grid of pictograms that a person points to or taps to say something. Unlike a sequence, it has no order: it is a vocabulary in plain sight, always available.

In **PICTOS.net**, each square on a board can have a background colour for its type of word and, if you like, a voice recording that plays when it is tapped. Like sequences, boards use pictograms you already have in the library.

Square colours

The colours follow the **Fitzgerald key**, a way of organising widely used in augmentative and alternative communication: each colour groups one type of word, so the person finds what they want faster and can build up sentences.[7]

	Colour	Name in PICTOS	Example words
	Yellow	**People**	I, you, Mum, teacher
	Green	**Verbs**	eat, go, help, have a shower
	Blue	**Adjectives**	happy, sad, big, cold
	Orange	**Nouns**	water, toilet, bus, pill
	Pink	**Functional**	and, with, in, because
	Purple	**Social**	kia ora, thanks, please, no thanks
	Grey	**System**	empty squares, ‘delete’, ‘next page’

[7]The Fitzgerald key has variations from country to country and between communication systems. If the person already uses other colours, keep to them: the colours in PICTOS.net are only an aid to organisation.

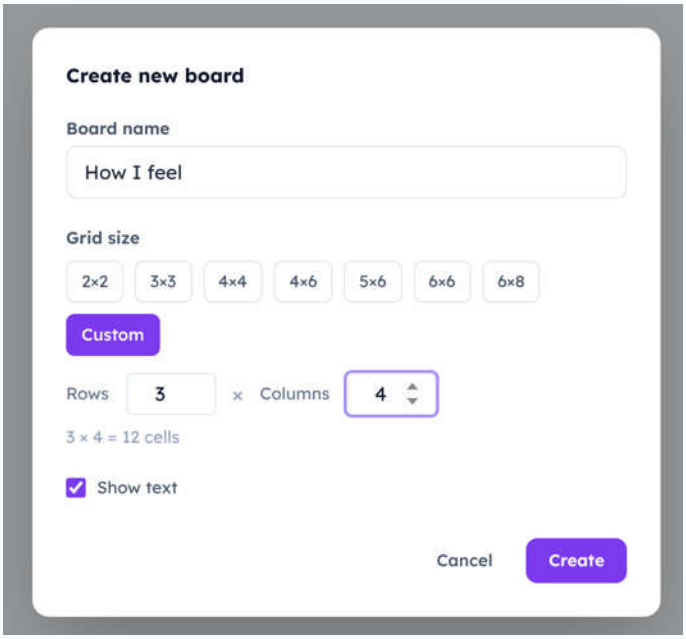
Example: a board for saying how I feel

We will build a board with three rows and four columns, covering needs, feelings and social phrases:

- **Row 1, green (actions and needs):** A bit peckish, Grab a coffee, Hit the sack, Chill out.
- **Row 2, blue (how I am):** Under the weather, Over the moon, Down in the dumps, In over my head.
- **Row 3, purple (social):** Give us a hand with this, Thanks a million, Give me a minute, Leave me be.

Create the board

Click the **Boards** tab and then the **Create new board** box. Type the name How I feel and choose a size. There are ready-made sizes (2×2, 3×3, 4×4, 4×6, 5×6, 6×6 and 6×8); for 3×4 choose **Custom** and enter 3 rows and 4 columns. Leave **Show text** ticked if you want the phrase under each pictogram.

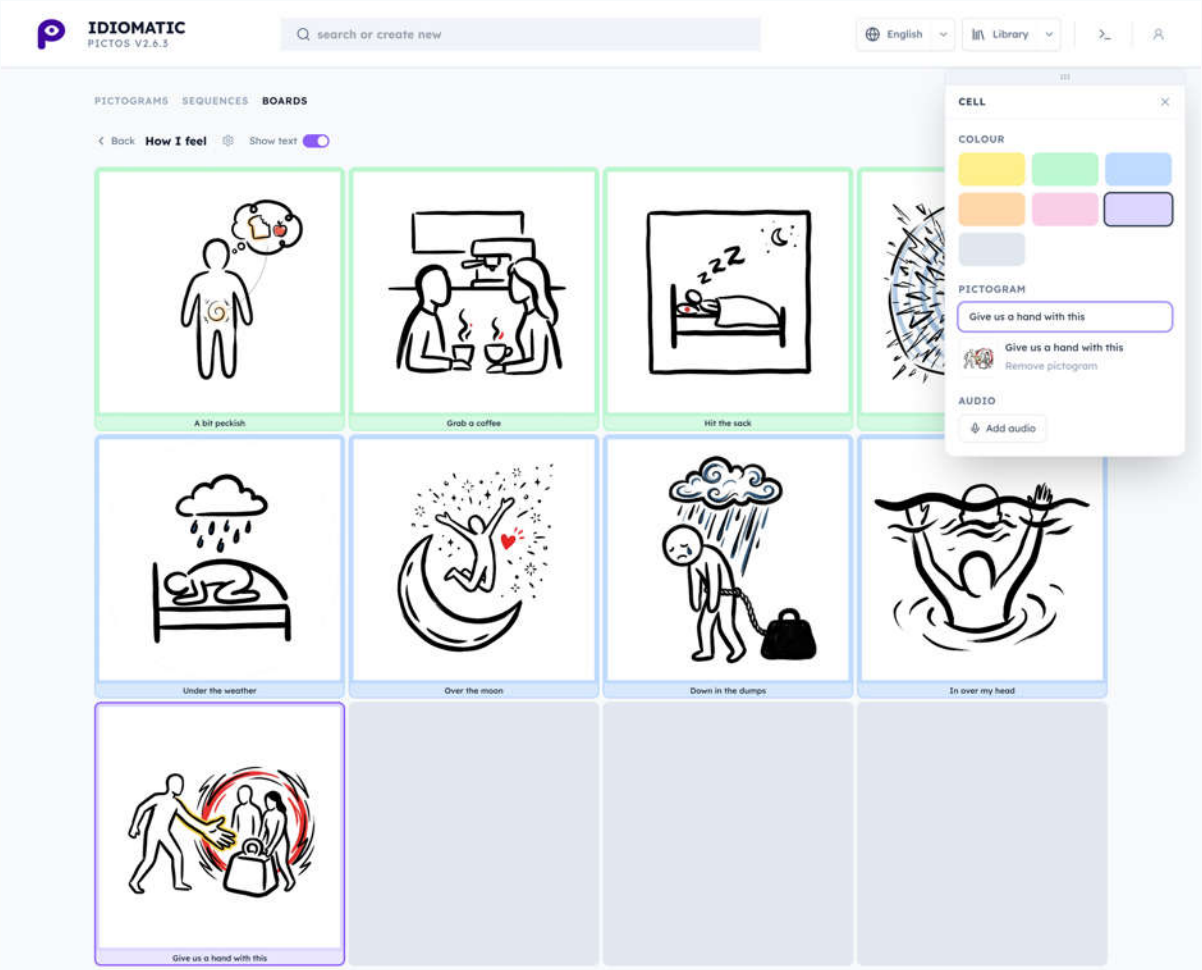


Creating a board: name, size and visible text.

Fill the squares

The board opens ready to edit (the pencil at the top right is selected), with every square grey. Click a square and the **Cell** box appears:

1. Under **Colour**, choose the shade. For example, green for ‘A bit peckish’.
2. Under **Pictogram**, type part of the phrase (‘peckish’) and click the phrase that appears underneath.
3. Close the box with the cross, or simply click the next square. If the box covers a square you want to see, move it by its top edge.

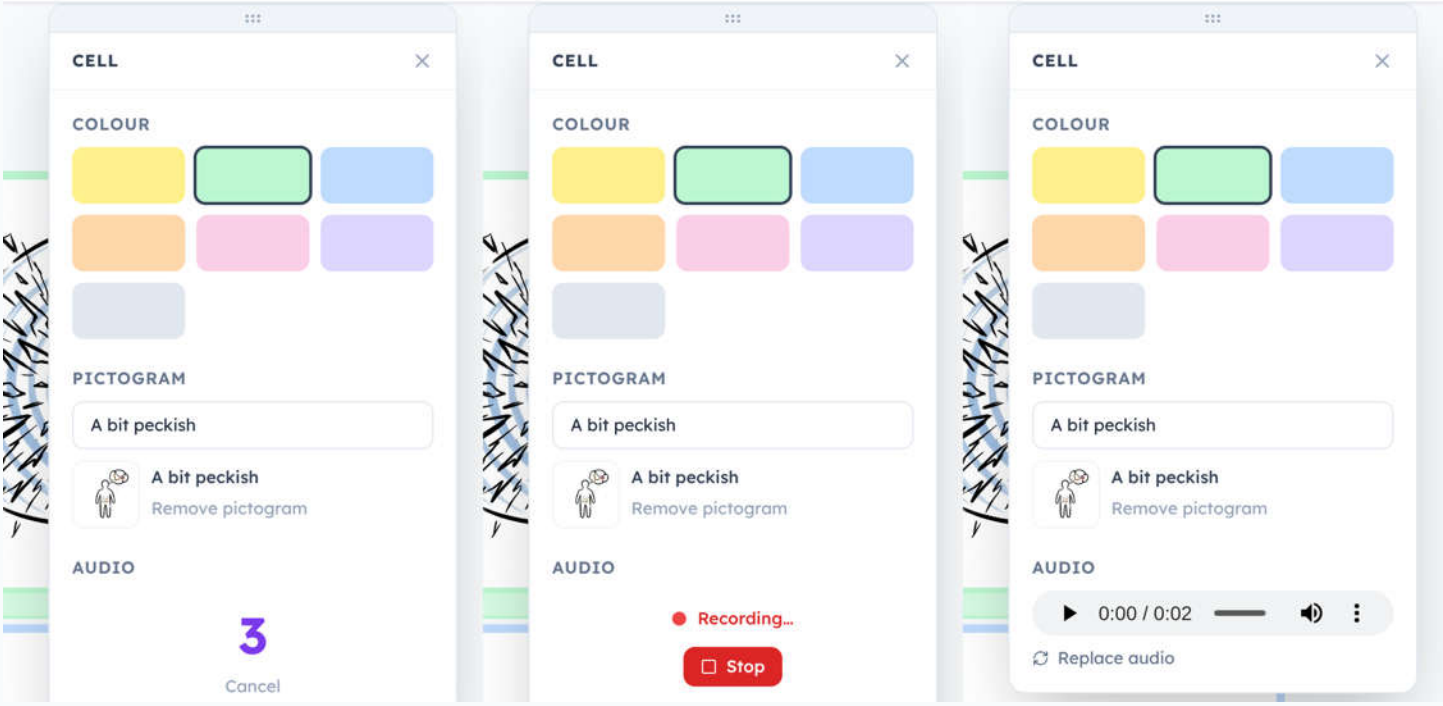


The Cell box: colour, pictogram and audio for the chosen square.

Record a voice

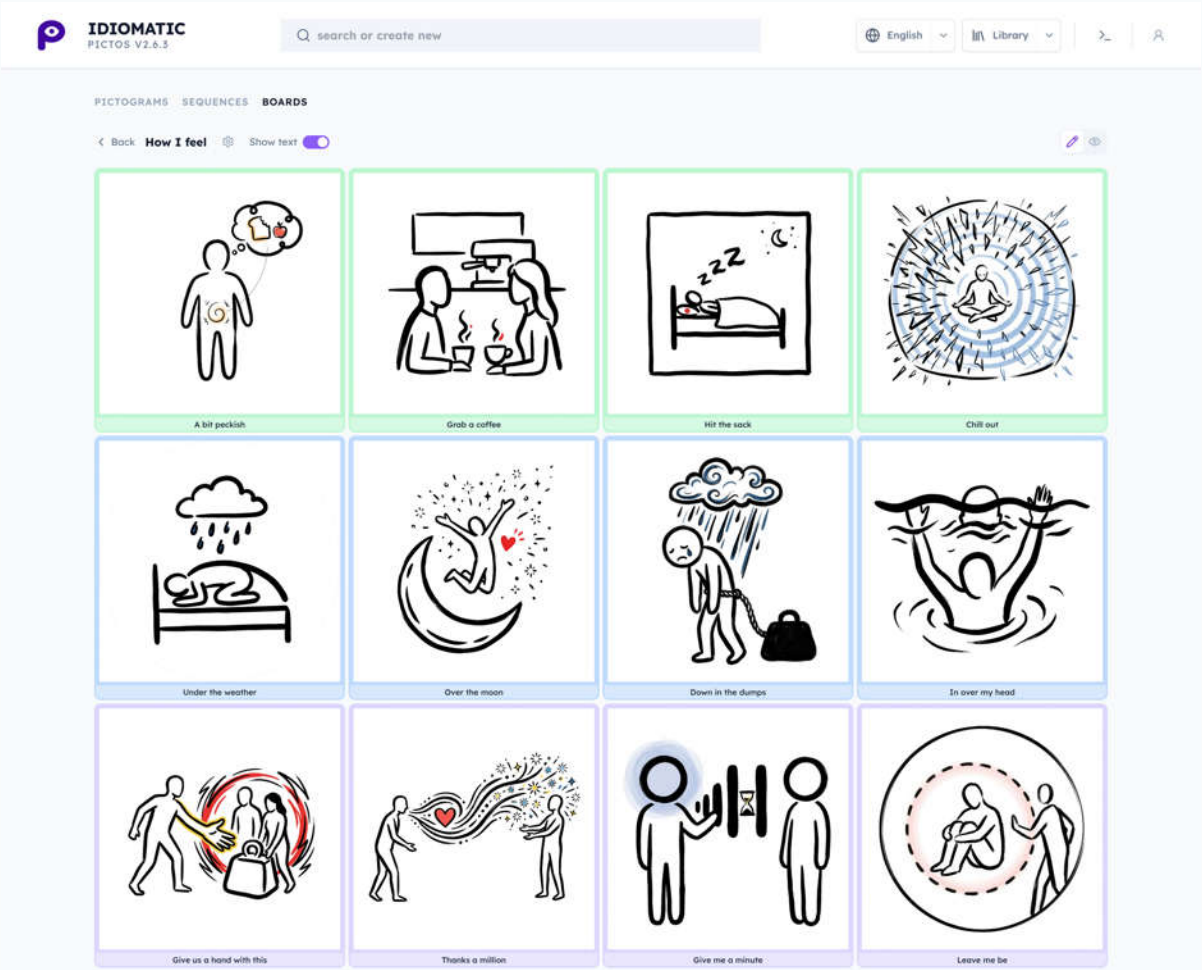
Every square with a pictogram can have a recording. Ideally it is recorded by someone close to the person, in the voice and accent of their everyday world. You need a computer with a microphone (nearly all laptops have one).

1. In the Cell box, click **Add audio**.
2. The browser will ask whether to allow the microphone: say yes.
3. A countdown appears: 3, 2, 1. When it finishes, say the phrase.
4. Click **Stop**. The recording is ready to play back. If you are not happy with it, click **Replace audio** and record it again.



Countdown, recording in progress, and the finished audio.

The recording belongs to the pictogram, not to the square. If ‘A bit peckish’ appears on other boards in the same library, they will all play the same recording.[8]



The finished ‘How I feel’ board.

[8] If the board will be used on an iPhone or iPad, a ‘Convert audio for iOS’ button appears, which adapts the recordings so they play on those devices.

Use the board

When the board is ready, click the **eye** icon at the top right. The board fills the whole screen, with no menus or distractions. Tapping a square with a recording plays the voice; squares without one do nothing. To leave, tap the small arrow at the top left or press the **Esc** key on the keyboard.



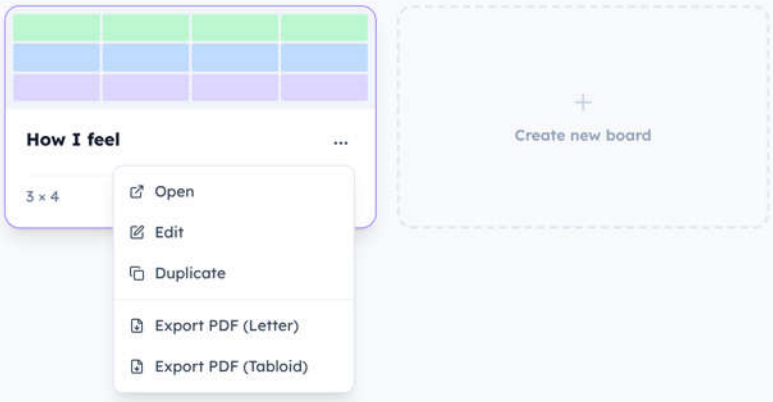
The board full screen, ready to tap.

On a laptop or tablet this view works as a simple communication device. It is worth trying it out with the person and adjusting the grid size to what they find comfortable to see and touch.

Print, duplicate and edit

In the Boards tab, each board has three dots '…' that open these options:

- **Open**: goes into the board.
- **Edit**: changes the name, the size or whether text is shown. **Delete** is here too. Careful: if you make the board smaller, the squares that fall outside it are lost.
- **Duplicate**: makes a copy, for example to prepare a version with fewer squares for someone else.
- **Export PDF (Letter)** and **Export PDF (Tabloid)**: prepare a sheet for printing with the whole board, its colours and its text, ready to laminate. Tabloid is a sheet twice the size of Letter, close to A3.



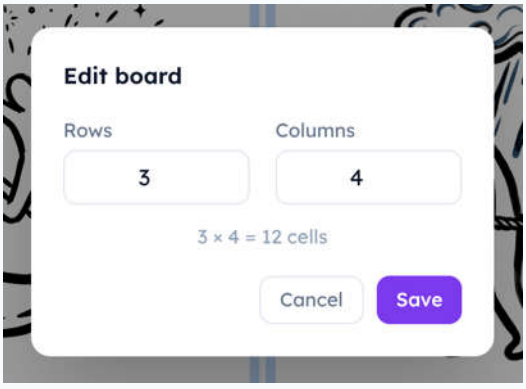
Options for a board.

This is the ‘How I feel’ board ready to print on Letter paper:



The board on a Letter sheet, ready to print.

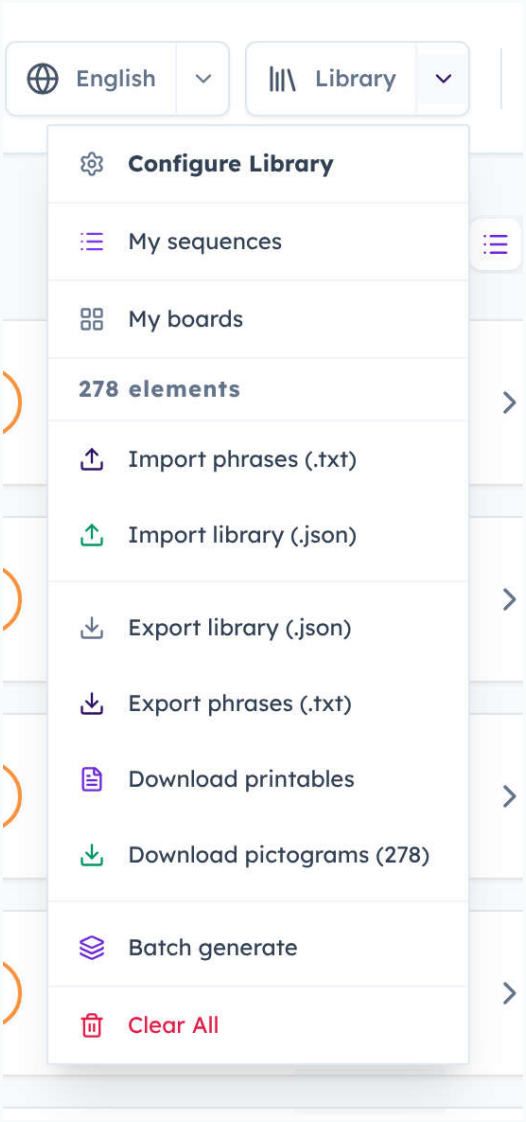
Inside the board, the gear icon next to its name lets you change the number of rows and columns quickly.



Changing the size from the gear icon.

The Library menu

The little arrow next to the **Library** button, at the top right, opens a menu with everything you can do with the library you have open:



The Library menu.

Configure Library. Opens the settings we looked at earlier.

My sequences and **My boards.** Take you straight to those tabs.

Import phrases (.txt). Loads a whole list of phrases at once from a plain text document (like the ones made in Notepad), one phrase per line.

Import library (.json). Opens a library someone sent you, or one you saved earlier. A library file ends in '.json'.

Export library (.json). Saves the whole library in a single file, with its sequences, boards and recordings. This is how you move it to another computer or share it with a colleague.

Export phrases (.txt). Saves just the list of phrases.

Download printables. Prepares a document for printing with every pictogram and its phrase, ready to cut out.

Download pictograms. Saves every drawing onto your computer, each one named after its phrase.

Batch generate. Draws, in one go, every pictogram that is waiting for its drawing. It is handy after importing a long list of phrases. It can take anywhere from minutes to a few hours, and the drawings arrive even if you close the browser.

Clear All. Empties the library. It asks you to confirm first.

Privacy and professional judgement

Your data stays on your computer

Everything you type, and every pictogram, sequence, board and recording, is saved on the computer you work on, inside the browser. None of it is kept on **PICTOS.net** servers.[9]

If you close **PICTOS.net** and come back days later, everything is still there. But note: on another computer or another browser you will not see your libraries. To take them with you, use **Export library** on the first one and **Import library** on the second. And to keep a backup of all your libraries together, use **Back up libraries** in the purple panel on the home screen.

Professional judgement always decides

The pictogram that **PICTOS.net** proposes is only ever a suggestion, never an instruction. The tool can get the cultural tone, the clinical detail or the way a situation is shown wrong.

That is why every pictogram can be redrawn, corrected, thrown away or replaced. The artificial intelligence suggests; the person who knows the individual they support decides.

[9]To create a pictogram, the phrase is sent to the artificial intelligence services that process it; the drawing comes back to your browser and is stored there. The details are in the site's privacy policy.

An invitation

I am **Herbert Spencer**, an interaction designer and doctoral researcher in cognitive accessibility (AUT University, Aotearoa New Zealand; affiliated with the Pontificia Universidad Católica de Valparaíso, Chile).

PICTOS.net is part of my research into how artificial intelligence can support professionals who work with autistic people and others with complex communication needs, without replacing them or taking away their authority over their own practice.

It is a tool designed to be reviewed, criticised and improved by professionals like you.

How to get in touch

I would love to hear how you get on: which phrases work, which do not, which drawing style feels right for each situation, what made you hesitate, and what you wish the tool did that it does not do yet.

At the bottom of every page of pictos.net there is a **Contact** link. You can also write to me directly at hspencer@ead.cl. If you would like to share a library, attach the file made with **Export library** to your email.

Thank you for reading this far, and for considering **PICTOS.net** in your practice.

Herbert Spencer González

http://herbertspencer.net